

NIRMA UNIVERSITY
MCA – Semester I
6CS103CC26 – OPERATING SYSTEM

QUESTION BANK

1. Differentiate between User mode and Kernel mode.
2. Explain the concept of interrupts and discuss why interrupts are required in a computer system.
3. Describe the various services provided by the operating system. Also, define the goals of operating systems.
4. What are the main components of a computer system? Explain their functions.
5. Define process states. Design a 5-state process model using a proper state transition diagram.
6. What is a Process Control Block (PCB)? Draw the diagram of PCB and explain each block of PCB in detail.
7. What is a thread? Explain the need for threads in an operating system.
8. Differentiate between Multiprogramming and Multiprocessing.
9. Explain the role of Main Memory and I/O Modules in a computer system.
10. What is a System Bus? Explain its purpose and the types of information transferred through it.
11. What is a process? Differentiate between a program and a process.
12. Explain the five-state process model and describe the New, Ready, Running, Blocked/Waiting, and Exit states.
13. What is cache memory? Why is it used in a computer system?
14. Differentiate between User-Level Threads (ULT) and Kernel-Level Threads (KLT).
15. What is mutual exclusion? Explain the requirements that a solution to the Mutual Exclusion problem should satisfy.
16. What is a critical section? Why is mutual exclusion required in concurrent processes?
17. Define deadlock. Explain the necessary conditions for deadlock.
18. Illustrate the Resource Allocation Graph (RAG) with a suitable diagram and explain how it can be used to identify deadlock.
19. What is a semaphore? Explain the basic idea of semaphore operations.
20. Differentiate between binary and counting semaphores.
21. Explain the role of the Processor in a computer system and describe the functions of processor registers.
22. What are Program Counter (PC) and Instruction Register (IR)? Explain their roles.
23. Explain the difference between Memory Address Register (MAR) and Memory Buffer Register (MBR).
24. What is kernel mode? Why is it more privileged than user mode?
25. Explain multiprogramming and discuss how it improves processor utilization.
26. What is the kernel of an operating system? Explain its role.

27. What is process switching? Explain the role of the program counter and processor registers during a process switch.
28. What events can cause a process to change its state? Explain with suitable examples.
29. What are User-Level Threads? Explain their advantages and limitations.
30. What are Kernel-Level Threads? Explain their advantages and limitations.
31. What is Symmetric Multiprocessing (SMP)? List its important characteristics.
32. What is a microkernel? Explain the basic idea and benefits of microkernel organization.
33. What is concurrency? Explain its relationship with mutual exclusion and synchronization.
34. What is starvation? Explain how it differs from deadlock in concurrent systems.
35. Explain the semWait and semSignal operations of a semaphore.
36. Differentiate between deadlock prevention, deadlock avoidance, and deadlock detection.
37. What is a safe state and what is an unsafe state in deadlock avoidance?
38. Explain the Banker's Algorithm for deadlock avoidance and describe the information required to determine whether a state is safe.